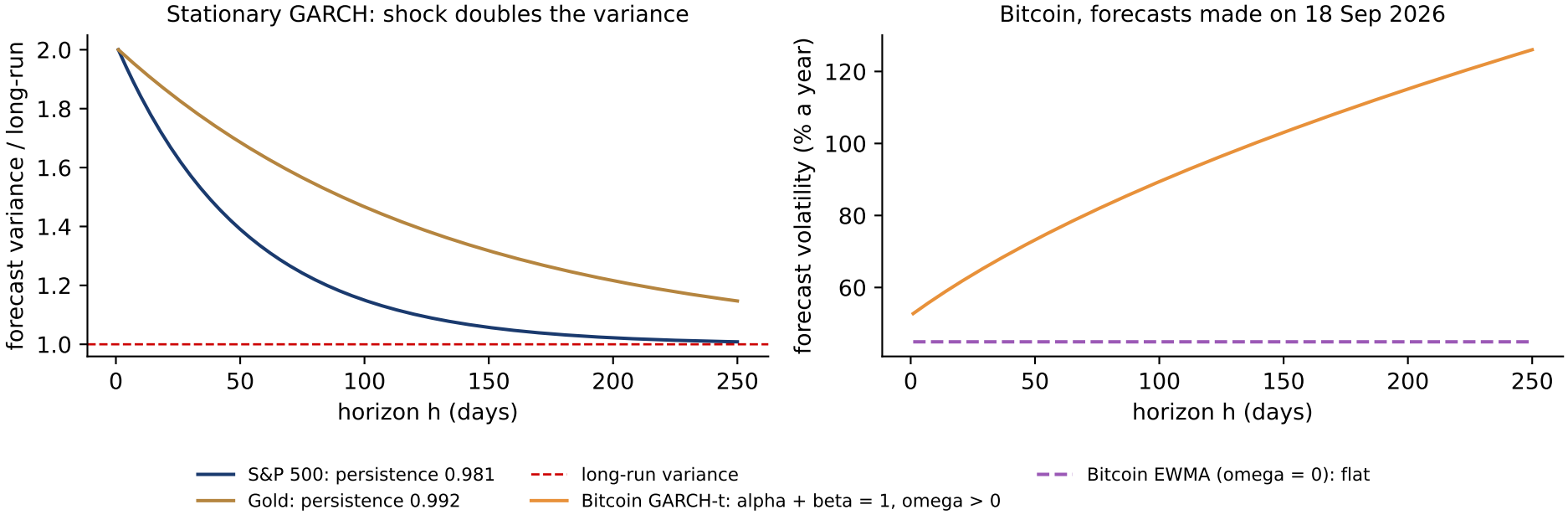
Stationary GARCH: shock doubles the variance
forecast variance / long-run
2.0
1.8
1.6
1.4
1.2
1.0
0
50
100
150
200
250
horizon h (days)
Bitcoin, forecasts made on 18 Sep 2026
forecast volatility (% a year)
120
100
80
60
0
50
100
150
200
250
horizon h (days)
S&P 500: persistence 0.981
Gold: persistence 0.992
long-run variance
Bitcoin GARCH-t: alpha + beta = 1, omega > 0
Bitcoin EWMA (omega = 0): flat